AI & Data Analytics Assignment

Objectives

The aim of the assignment is to use **Power BI** to:

- Perform **data extraction, cleansing and transformation** to pre-process data prior to analysis.
- Perform **data exploratory analysis** to identify underlying data patterns, trends and analytical insights.
- Develop **interactive dashboards** with a global perspective to address international and cultural differences, diverse needs and expectations of the key stakeholders.

In addition, you will be exposed to **AI Fluency** by describing AI application, value, limitations and societal impact.

Deliverables

The assignment submission is due on **Week 16 Sun 2359 (8 Feb 2026)**.

The Assignment comprises 35% of the total Learning Unit. Refer to *Assessment Rubrics*.

Group Submission (10%)

1. Word/PDF document on **Defining data needs** (max 2 pages). Refer to *Annex A – Concise example of Group submission.*
2. **Integrated** Power BI Desktop file (.pbix) with all data sources used.
3. Published Power BI dashboard link (if applicable).

Individual Submission (25%)

1. Word/PDF document on **Data analysis & visualization** (max 5 pages). Document by illustrating with screenshots and code snippets (if applicable). Refer to *Annex B – Concise example of Individual submission.*
2. Anthrophic AI Fluency Certificate.

Requirements

Referring to the *Business Scenario*, define the **data needs**, cleanse, explore and **analyse the data** to create a **data visualization report** with **data-driven insights** to influence stakeholders.

Form a group of 3-4 students. Using Power BI, each group is to create a data visualization presented in an integrated dashboard. Each student is required to choose a focus area for analysis, creating at least 4 visuals in the Power BI dashboard. Design the dashboard with intuitive user experience and navigational links to showcase each member's reports. Using effective data-driven storytelling, present your data visualization, focusing on the insights.

Tasks

1. Define Data Needs (Group)

- The **audience** who will use the visualization
- The **purpose** of the visualization
- The **questions** that need to be answered
- The **metrics and KPIs** to be visualized
- The **data sources** required

As a group, define the data needs before starting with the data collection. Defining data needs is an **iterative** process. As you explore the data, your group might discover new questions that needs to be answered or new insights waiting to be uncovered. You may use Gen AI to help you find focus areas and explore metrics and KPIs. However, the data sources should be real data and not artificially generated by AI to ensure that your perspective is real and authentic. Do note that irrelevant use of datasets may impact on the performance of your report.

Document all data sources in a **table** by providing the specific **URL link**.

Market size of sustainable fashion	Secondhand apparel market value by segment worldwide 2019-2029| Statista
Singapore Retail Sales Index	Retail Sales Index, (2017 = 100), In Chained Volume Terms, Annual

You may use any datasets from reliable sources be it local or global to support your analysis.

- Statista
- Kaggle
- Data Gov
- Singapore Statistics
- Our World in Data
- World Bank Open Data
- OECD Better Life Index
- ... any other appropriate data set

2. Data Preparation & Exploration

Data Preparation

Prepare the raw data. You may also use AI features in Power BI to help you with the data prep.

a) **Data Cleansing**
- Handle missing values
- Remove duplicates
- Correct errors and inconsistencies
- Identify outliers

b) **Data Structuring**
- Model or organize data
- Set column names, data types and constraints
- Flaten nested or hierarchical data if needed
- Establish keys and joins to link related data tables

c) **Data Transformation**
- Normalization or scaling of features
- Encoding categorical variables (e.g., one-hot encoding, label encoding)
- Aggregation or summarization
- Creating new features (feature engineering)
- Data type conversion

Data Exploration

You many consider the following data exploration techniques to evaluate whether there's real insight that can be uncovered through data analysis.

a) **Descriptive and Exploratory Analysis** → Understand the data
   - Summarizing the main features of the data using statistics (mean, median, mode, variance).
   - Visualizing data distributions, central tendencies, and variability.
   - Identifying trends and patterns.
   - Time-series analysis
   - Identifying relationships between variables such as correlation analysis

b) **Inferential and Statistical Modeling** → Test hypotheses and generalize findings
   - Drawing conclusions about populations from samples.
   - Using visual and statistical techniques to generate hypotheses.
   - Estimating parameters and confidence intervals.
   - Drawing conclusions about populations from samples.
   - Estimating parameters and confidence intervals.

c) **Predictive and Multivariate Analysis** → Forecast and understand complex relationships
   - Using statistical models or machine learning algorithms to predict future outcomes.
   - Building regression, classification, or clustering models.
   - Analyzing more than two variables simultaneously
   - Understanding complex interactions and patterns.

3. Data Analysis & Visualization

After data exploration, summarize the insights to help the target audience interpret the results of the data analysis with visualizations.

You may consider the following types of data analysis.

a) **Descriptive Analysis** → What happened (Past)
   Descriptive analysis provides an easily digestible snapshot of what has happened in the past. One of the biggest uses of descriptive analysis in business is to track KPIs (key performance indicators) to describe how a business is performing on all levels. It can be used to track your marketing KPIs or CEO KPIs, everything that you need and should be keeping a close eye on.

b) **Diagnostic Analysis** → Why did it happen? (Past)
   Diagnostic Analysis helps users gain a contextual understanding of why something has happened. It is characterized by techniques such as drill-down, data discovery,

data mining, and correlations. Businesses perform this type of data analysis to understand the connections between data and identify patterns of behavior.

c) **Predictive Analysis** → What is likely to happen? (Future)
Predictive Analysis estimates the likelihood of future events and outcomes. This is especially useful for businesses as it enables them to plan ahead and make strategies based on facts. The predictive analysis involves analytic techniques such as regression, classification, outliers, time series analysis, and more.

d) **Prescriptive Analysis** → How will it happen? (Future)
Prescriptive Analysis shows you how you can best take advantage of the future outcomes that have been predicted and answer the question of how you will make it happen. It is the most difficult analysis to perform.

Include **screenshot** of each **Power BI page** and describe its **purpose**. Each page should have a title, and can contain a few visuals, together with filters. Your visualization should not just merely present data. The visuals should unravel data to reveal **insights** that help the target audience make data-driven decisions. Add **interactivity** to your dashboard.

4. Data Storytelling & Presentation

Present to your tutor the final integrated Power BI dashboard. Use data analysis, visualizations, and a compelling narrative to effectively communicate complex data-driven insights to your target audience, ultimately leading to better understanding and decision-making.

Timeline

Before the final submission, there are some checkpoints that you can consult with your tutor by showing your progress and deliverables. Your tutor can help fine-tune your assignment by providing feedback and suggestions. Do **NOT** wait till the last minute to work on your deliverables.

Week	Task	Checkpoint
2	Form group	
3	Define data needs	
4	Collect data	Documentation of data needs
5 - 7	Clean and explore data	

8 - 9	Analyse and visualize data	Documentation of data cleansing, analysis and visualization
13	Implementing ideas effectively	
15	Technical walkthrough	Demo your technical expertise, such as the usage of AI tools, formulas, script etc.
16	Presentation	Final presentation & submission

Business Scenario

Background:

A Singapore-based sustainability research center is developing a data-driven report to evaluate whether **Singapore is a sustainable place to live in**. The goal is to understand the country's long-term livability by analyzing multiple sectors such as healthcare, education, housing, safety, environment, and cost of living. The study will benchmark Singapore against global standards and regional peers to provide evidence-based insights on its sustainability and attractiveness as a place of residence.

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Your group can choose to emphasize one anchor sector such as healthcare, housing, or environment, or analyze across multiple domains for a holistic view. The research center is fictitious and aims to use data storytelling and Power BI dashboards to communicate findings to policymakers, businesses, and the public.

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The organization wants to use **data-driven insights** to:

- Evaluate Singapore's performance across key **social and economic sectors**.
- Analyse the **nation's progress in environmental sustainability.**
- Compare overall **affordability and quality of life** with other global locations
- Identify **strengths and challenges** in achieving sustainable development.

Objectives:

- **Benchmark Singapore's liveability** and **sustainability** against international and regional standards.
- **Conduct in-depth analysis** of national-level data across multiple dimensions.
- **Evaluate the balance** between affordability and overall quality of life.
- **Generate recommendations** and **insights** to strengthen long-term sustainability.

Focus Areas: (not limited to)

Sustainability & Liveability Scorecard

- Use multi-sector data (health, housing, education, environment, safety).
- Develop composite indicators such as *Sustainability Index* or *Liveability Score*.

Cost of Living vs. Quality-of-Life Trade-Off

- Compare housing affordability, transportation costs, and income vs. global cities.
- Financial metrics: Household expenditure share, healthcare spending per capita, education affordability.

Environmental Sustainability

- Measure renewable energy share, CO_2 emissions, and waste management efficiency.
- Compare trends with regional peers and developed nations.

Long-Term Forecasting

- Project outcomes of population growth, climate challenges, and urban housing demand.
- Assess whether Singapore can remain one of the world's most sustainable and liveable cities.

Assessment Rubrics

Not competent	Developing	Functional	Competent	Proficient	Weightage
1. Define Data Needs – Group					5%
0m to < 2.5m	2.5m to < 3m	3m to < 3.5m	3.5m to < 4m	4m to 5m	
Data needs are unclear or poorly defined. Key metrics/KPIs are missing or irrelevant. Data sources are inappropriate. Many details are missing, and important questions are ignored.	Data needs are vague. Key metrics/KPIs are ambiguous. Data sources lack justification. Some important questions are poorly addressed.	Data needs lack measurable detail. Key metrics/KPIs are partially described. Data sources are poorly matched. Some important questions are not anticipated.	Data needs are mostly clear.. Most important metrics/KPIs are identified. Data sources are mostly justified. Relevant questions are considered.	Clear data needs covering key metrics and KPIs, including well-justified data sources. Address questions by anticipating all details and constraints.	
2. Dashboard – Group					5%
0m to < 2.5m	2.5m to < 3m	3m to < 3.5m	3.5m to < 4m	4m to 5m	
Unintegrated dashboard with no interactivity and navigation. Poor design choices showing just data without meaning.	Partially integrated dashboard with limited interactivity. Poor navigation and inconsistent design with shallow insights.	Mostly integrated dashboard with slight interactivity. Some usability problems and unpolished design with some insights that could go deeper.	Integrated dashboard with some interactivity, mostly easy to use and well-designed. Mostly insightful.	**Integrated** and **interactive** dashboard with **intuitive navigation** and **consistent theme**, showcasing **clear actionable insights**.	

Not competent	Developing	Functional	Competent	Proficient	Weightage
3. Data Preparation & Exploration – Individual					5%
0m to < 2.5m	2.5m to < 3m	3m to < 3.5m	3.5m to < 4m	4m to 5m	
Minimal or no cleaning or transformation. Missing data and outliers ignored. Data sources not combined at all. Investigates and review data/information gathered lacking data-driven support, often using guess work to make choices.	Some cleaning attempted. Limited transformation applied. Missing values and errors not addressed. Data poorly formatted and inconsistent. Integrated data sources with errors. Investigates and review data/information gathered in a superficial manner, often lacks robust data evaluation to make choices.	Basic cleaning and transformations. Most missing values handled with few errors or duplicates. Data sources integrated with minor omissions. Investigates and review data/information gathered in a basic manner, so as to make choices, though may not fully integrate all data.	Mostly clean and transformed data with minor issues. Missing data handled comprehensively. Data integrated with well-managed merges. Investigates and review data/information gathered fairly, so as to make informed choices with occasional insights.	Data is thoroughly **cleansed** and **transformed**. **Missing values and outliers** are properly handled. Multiple **data sources merged** seamlessly. Consistently **investigate and review complex data/information** gathered fairly and in detail, so as to make well-substantiated choices with **deep insights**.	
4. Data Analysis & Visualization – Individual					10%
0m to < 5m	5m to <6m	6m to < 7m	7m to < 8m	8m to 10m	
Visuals are misleading, with missing labels and difficult to read.	Visuals are confusing and cluttered. Most are	Visuals are mostly clear with minor readability issues.	Visuals are clear and mostly relevant, but some could be optimized.	Correct visual used. Visuals are **clear, well-labelled, easy to interpret**, and	

Not competent	Developing	Functional	Competent	Proficient	Weightage
	irrelevant or misleading.	Some are off-topic and redundant.		highly **relevant to key metrics and goals**.	
5. Data Storytelling – Individual					5%
0m to < 2.5m	2.5m to < 3m	3m to < 3.5m	3.5m to < 4m	4m to 5m	
Shares opinions that are often unsupported or irrelevant, with little to no logical structuring of evidence.	Shares opinions that are often not supported by relevant evidence, and lacking clarity and coherence	Shares opinions that are generally supported by evidence but occasionally lacks clarity or relevance.	Expresses clear opinions supported by logical evidence and reasons, though may lack depth.	**Articulates well-structured opinions** with **compelling, logically structured evidence** and rationale.	
6. AI Fluency – Individual					5%
0m	5m			10m	
No certificate submission. Did not attempt eLearning.	Did not submit certificate but attempted eLearning with screenshot of completion percentage as evidence.			Submit **AI Fluency certificate**	
				TOTAL	35%

Annex A – Concise example of Group submission

Define Data Needs

1. The **audience** who will use the visualization
   Venture capitalists who will invest in the business.

2. The **purpose** of the visualization
   - To provide clear, actionable insights on **which global markets to target** for expansion
   - To **compare market demand** and consumer behavior across regions
   - To **highlight economic and cultural factors** affecting sustainable fashion adoption
   - To **evaluate financial risks and opportunities** for each potential market
   - To support **data-driven decision making** for product localization, pricing, and supply chain planning

3. The **questions** that need to be answered
   - Which international markets show the highest demand for sustainable fashion?
   - How does consumer spending on apparel vary by region?
   - What are the key economic indicators (e.g., GDP per capita, disposable income) relevant to market potential?
   - How do cultural preferences impact product style and sustainability perception?
   - What are the estimated market entry costs and expected revenues per region?
   - What pricing strategies maximize profitability given regional economic diversity?
   - How can supply chains be optimized for cost efficiency and sustainability in each region?

4. The **metrics and KPIs** to be visualized
   - **Market demand indicators:** Search trends, sales volume of sustainable fashion, growth rates by region
   - **Consumer spending metrics:** Average apparel expenditure per capita, income segmentation

- **Economic KPIs:** GDP per capita, inflation rate, unemployment rate, purchasing power parity (PPP)
- **Financial metrics:** Market entry cost estimates, projected revenue, profit margins, ROI
- **Operational KPIs:** Supply chain costs, delivery lead times, pricing elasticity, inventory turnover
- **Cultural engagement:** Customer satisfaction scores, product preferences, sustainability awareness index

5. The **data sources** required
   - **Market data:** Industry reports, fashion sales data, Google Trends, e-commerce platform analytics
   - **Economic data:** World Bank, IMF, government statistical agencies
   - **Consumer behavior data:** Market surveys, social media sentiment analysis, customer feedback
   - **Financial data:** Internal cost estimates, competitor pricing, financial statements
   - **Supply chain data:** Logistics providers, inventory management systems, regional tariffs and regulations
   - **Cultural data:** Market research studies, demographic profiles, lifestyle reports

Annex B – Concise example of Individual submission

Data Preparation, Exploration & Analysis

Include all DAX formulas and scripts you have used.

Market Segmentation:

- Segment countries by demand indicators, market maturity, and consumer awareness.

Trend Analysis:

- Use time series data to find emerging markets.

Descriptive Statistics:

- Average apparel spend per capita, per age group, per region.

Clustering:

- Group countries by spending patterns (e.g., low, mid, high spenders).

Correlation Analysis:

- Between GDP per capita and sustainable fashion spending.

PCA (Principal Component Analysis):

- Reduce cultural/economic factors to key driving dimensions.

Data Analysis & Visualization

Include screenshots of Power BI pages.

1. **Executive Summary**
   - Global map with market ratings
   - KPI scorecards (e.g., CAC, ROI, revenue potential)
   - Filters by region, demographic, scenario
   - Charts tracking cost, revenue, and profitability over time

2. **Identify High-Potential Markets for Sustainable Fashion**

   Purpose: Determine which countries/regions have strong demand and growth potential for sustainable apparel.

   Visuals:

- **Heat Map**:
  - Shows demand intensity for sustainable fashion across countries.
  - Can be based on search volume, market size, or sustainability awareness index.
- **Bubble Chart (Market Attractiveness vs. Ease of Entry)**:
  - X-axis: Ease of market entry (regulations, logistics).
  - Y-axis: Market attractiveness (size, growth).
  - Bubble size: Market size for sustainable apparel.
- **Bar Chart (Top 10 Markets by Market Size)**:
  - Compare sustainable fashion markets (e.g., Germany, US, Japan, South Korea).

3. **Understand Consumer Spending Behavior on Apparel by Region**

   Purpose: Analyze how much people spend on apparel and how that varies by region.

   Visuals:

   - **Stacked Bar Chart**:
     - Apparel spending by category (fast fashion, premium, sustainable) across regions.
   - **Line Graph**:
     - Trends in consumer apparel spending over the past 5–10 years.
   - **Pie Charts** (by Region):
     - Distribution of consumer spending on different fashion categories (e.g., essentials vs. luxury).
   - **Histogram**:
     - Consumer spending per capita on apparel in selected markets.

4. **Build a Financial Model for Market Entry Costs vs. Projected Revenue**

   Purpose: Visualize the financial feasibility of entering different markets.

Visuals:

- **ROI Heatmap**:
  - Projected Return on Investment by country.
- **Cost vs. Revenue Line Graph**:
  - Cumulative cost vs. revenue over time for each market.

5. Optimize Pricing and Supply Chain to Maximize Profitability

Purpose: Find the best pricing strategy and supply chain model by region.

Visuals:

- **Price Sensitivity Curve**:
  - Visualize consumer response to price changes.
- **Break-even Analysis Chart**:
  - Determine at what volume or price the company becomes profitable in each market.
- **Stacked Area Chart**:
  - Compare shipping/logistics costs per region over time.